

Bilal Mahmood

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Expected May 2028

Bachelor of Science in Mechanical Engineering, Minor in Business

Relevant Coursework: Statics and the Mechanics of Solids, Thermodynamics, Intro. to Computing for Engineers

WORK & TECHNICAL EXPERIENCE

AutoBoat Project Team

Ithaca, New York

Mechanical and Manufacturing Subteam

Jan. 2025 - Present

- CNC-fabricated foam-core female molds and executed fiberglass layups for ama hulls, reducing prototype mass by ~15% while improving repeatability.
- Conducted CFD and FEA simulations in ANSYS/Orca, identifying structural vulnerabilities and achieving a projected 10% improvement in hydrodynamic efficiency.
- Optimized material and adhesive selection (HDPE, plywood, King Starboard, epoxy systems), improving bonding strength and reducing rework by 20%.
- Standardized manufacturing procedures, creating scalable build documentation for future cycles.
- Integrated thruster mounts and hull architecture, from CAD through pool testing, validating propulsion stability and structural reliability.

Sure Shot Pest Control

Queens, New York

Service Representative

Jun. 2025 - Aug. 2025

- Coordinated service requests for 100+ clients, generating accurate work orders for dispatch to 20+ technicians.
- Strengthened problem-solving and communication skills through 10+ daily client interactions in a fast-paced environment, directly contributing to assessments made by upper management.

Modeling Elliptical Galaxies

Princeton, New Jersey

Research Assistant

Jun. 2023 - Sept. 2023

- Shadowed Princeton graduate student with research on anomalies in space-time modeling, focusing on modeling mass distributions in elliptical galaxies with Python, and how black holes can result in skewed models.
- Synthesized findings from Stephen Adler's work on star formation near a 'leaky' black hole to refine modeling assumptions, improving the validity of simulation results.
- Applied Mathematica for numerical analysis and LaTeX for technical documentation, enhancing technical fluency and research capabilities.

LEADERSHIP EXPERIENCE

Future for Young Scientists

Remote

Board Member/Educator

Jun. 2022 - Present

- Designed and delivered 20+ STEM lessons, from engineering to ecology, reaching 100+ students globally.
- Led 10+ outreach initiatives, including sponsor communications, event coordination, and performing live science demonstrations at 5+ community events, enhancing public engagement with hands-on learning.
- Expanded reach across 5+ countries and 10+ U.S. states through scalable online programs.

Science Olympiad

Hicksville, New York

President

Sept. 2018 - Jun. 2024

- Implemented a new team selection framework for 85+ applicants, streamlining roster creation and enhancing competitive readiness, improving team performance, and yielding the best state results in 12 years.
- Secured \$2,000+ in funding, fully covering travel and competition expenses for regional and state tournaments.
- Organized mentorship program for 60+ students, fostering peer learning and sustained team performance.

ADDITIONAL INFORMATION

Skills: Autodesk Fusion 360, Solidworks, ANSYS, Orca, Microsoft Office Suite (Excel, PowerPoint, Word), Google Workspace, Python, CAD, LaTeX, Mathematica | **Interests:** Billy Joel, Bollywood, H.I.M.Y.M, the New York Yankees, Trombone

Languages: English (Native), Urdu (Native), Hindi (Native), Punjabi (Conversational), French (Conversational)

Soft Skills: Leadership, teamwork, problem-solving, clear communication, time management, adaptability, optimism